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LIPA CITY

COLLEGE OF ENGINEERING
& ARCHITECTURE

Case Study: Enterprise Routing & Resilience for "Agri-Track Pilipinas"

Scenario Overview

Agri-Track Pilipinas is an agricultural technology firm with its central headquarters (HQ) in Manila and a critical data-collection branch located in Padre Garcia, Batangas. The Padre Garcia branch is responsible for real-time syncing of local livestock tracking data to the Manila HQ servers.

You have been brought in as the Lead Network Engineer. The company operates a dual-stack (IPv4 and IPv6) environment. They currently use a primary fiber connection to their ISP but have recently installed a secondary, lower-bandwidth cellular backup link at the Padre Garcia branch due to occasional fiber cuts in the province.

Your objective is to analyze the routing architecture, propose strict static routing configurations for the branch, and develop an incident response plan for a simulated outage.

Part 1: Architectural Design & Forwarding Logic

1. Packet Forwarding Optimization

The routers at the Manila HQ process thousands of data packets per second from various regional branches.

- **Task:** The company is currently using older legacy equipment that relies on process switching, where the CPU matches the destination address with an entry in the routing table for *every single packet*. Propose an upgrade to a modern packet forwarding mechanism. Explain how Cisco Express Forwarding (CEF) utilizes the Forwarding Information Base (FIB) and adjacency tables to forward packets more efficiently without constant CPU intervention.



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Answer:

The company must upgrade to Cisco Express Forwarding (CEF). In process switching, the router CPU checks the routing table for every single packet that arrives because of the bottleneck's routers. This causes high CPU usage and slower performance, especially when the Manila HQ router handles thousands of packets per second from branch offices. CEF eliminates this overhead by decoupling the data and control planes—pre-computing and storing forwarding paths in hardware tables before traffic even arrives.

The Cisco Express Forwarding (CEF) improves packet forwarding by using two important tables called the Forwarding Information Base (FIB) and the adjacency table. The FIB copies the routing table's best paths, letting the router check destinations instantly instead of querying the CPU for every packet. While the adjacency table stores Layer 2 details like destination MACs and outgoing interfaces. By handling these lookups in advance, the router forwards packets immediately without CPU overhead, ensuring a faster, more stable, and more efficient network capable of handling heavy real-time data loads (Cisco,n.d.).

2. Route Selection and Longest Match The Manila HQ router receives a packet destined for the IPv4 address 172.16.0.10. Its routing table contains three static routes: 172.16.0.0/12, 172.16.0.0/18, and 172.16.0.0/26.

- **Task:** Detail the router's path determination process. Explain the concept of the "longest match" and identify which of the three routes the router will select and why .



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Answer:

When The Manila HQ router receives a packet destined for the IPv4 address 172.16.0.10 it determine the best path by evaluating three available static routes: 172.16.0.0/12, 172.16.0.0/18, and 172.16.0.0/26. A router's path determination process follows a strict structural hierarchy, evaluating the longest prefix match first, followed by administrative distance, and finally the routing metric.

The router follows the rule called the “longest match”. This means the router will choose the route with the most specific subnet mask because it provides the most exact destination path. The router selects 172.16.0.0/26 for a destination of 172.16.0.10 because it has the longest subnet mask among the matching options. Under the longest-match rule, the router always prefers the most specific route available (Cisco,n.d.).

3. Branch Office Static Routing

The Padre Garcia branch router needs to be configured to send all unknown outward traffic to the ISP.

- **Task:** Formulate the exact command syntax required to implement a default static route (the Gateway of Last Resort) for both IPv4 (0.0.0.0 0.0.0.0) and IPv6 (::/0) .

Answer:

For IPv4, the router uses the command “ip route 0.0.0.0 0.0.0.0 <next-hop-address>” to create a default static route. This command tells the router to forward all unknown IPv4 traffic to the ISP through the specified next-hop address. It serves as Gateway whenever the router does not find a more specific route in the routing table (Cisco,n.d.).



- **IPv6 Link-Local Constraint:** For the IPv6 route, the ISP has provided a link-local address for the next hop. Explain why you must configure a "fully specified static route" (including both the exit interface and the next-hop IPv6 address) rather than just a next-hop route in this specific scenario.

Answer:

For IPv6, the command used is "ipv6 route ::/0 <exit-interface> <link-local-address>". This command allows the router to forward all unknown IPv6 traffic to the ISP using the specified exit interface and link-local next-hop address. It acts as the default route for IPv6 communication (Cisco,n.d.).

4. Implementing Link Resilience

To ensure the Padre Garcia livestock data never stops syncing, you must utilize the secondary cellular link.

- **Task:** Design a floating static route for this backup connection. Explain how manipulating the Administrative Distance (AD) allows this route to remain hidden in the routing table while the primary link (which defaults to an AD of 1) is active, and how it assumes control if the primary link fails .

Answer:

Padre Garcia branch should use a floating static route for the backup cellular connection to maintain continuous livestock data. A floating static route is a backup route. It only becomes active when the primary route fails. The backup route is configured with a higher Administrative Distance (AD). The primary static route uses the default AD of 1. The backup cellular route uses a higher AD value. This keeps the backup route hidden during normal operation. If the primary fiber link fails, the main route is removed from the routing table. The floating static route then becomes active automatically. This ensures continuous network reliability and uninterrupted traffic flow.



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Part 2: Incident Response & Troubleshooting

Incident: At 2:00 PM, the primary fiber link at the Padre Garcia branch goes down. However, the branch staff reports that they still cannot reach the Manila HQ servers, indicating the floating static route did not seamlessly take over.

1. Packet Processing Failure

- **Task:** Walk through the router's internal logic during this failure. When a packet from a local branch PC arrives at the router's interface, how does the router process it? Describe the steps the router takes when it decapsulates the packet, searches the routing table, fails to find a matching specific or default route, and ultimately drops the packet. What exact ICMP message is sent back to the source PC?.

Answer:

When the primary fiber link fails and the floating static route does not activate, the router cannot find a valid path to the Manila HQ servers, so the packet is dropped. The router then sends an ICMP Destination Unreachable message back to the source PC to inform it that the destination network cannot be reached (Cisco,n.d.).

- The router receives the packet from the local branch PC.
- It removes the Layer 2 header to read the destination IP address.
- The router checks the routing table for a matching route.
- It first searches for specific routes, then checks for a default route.
- Since the backup floating static route did not activate, no valid route is found.
- The router drops the packet instead of forwarding it.
- An ICMP Destination Unreachable message is sent back to the source PC.



The exact ICMP message sent back to the source PC is **ICMP Destination Unreachable**. This message informs the source device that the router cannot forward the packet because there is no available route to the destination network (Cisco,n.d.).

2. Troubleshooting Methodology

- **Task:** Develop a step-by-step troubleshooting matrix for the junior engineers on-site. Define exactly when and why they should use the following commands to isolate the fault:
 - ping and traceroute to verify Layer 3 connectivity and pathing .
 - show ip route to examine if the floating static route successfully populated the routing table .
 - show ip interface brief to check the operational status of the backup cellular interface .
 - show cdp neighbors to validate Layer 1 and Layer 2 connectivity to the ISP's edge equipment .

Answer:

- **ping** – Use this command to check if the router can successfully communicate with other devices and verify basic Layer 3 connectivity.
- **traceroute** – Use this command to identify where the connection path stops and determine which network hop is causing the problem.
- **show ip route** – Use this command to confirm if the floating static route appeared in the routing table after the primary link failed.
- **show ip interface brief** – Use this command to check whether the backup cellular interface is active and operating properly.
- **show cdp neighbors** – Use this command to verify that the router still detects and connects to the ISP's neighboring device at Layer 1 and Layer 2.



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